THE LONDON SCHOOL OF ECONOMICS AND POLITICAL SCIENCE

Summer 2022 Exam

GY462

Real Estate Finance

Suitable for all candidates

Instructions to candidates

This paper contains Two questions. Answer BOTH questions. All questions carry EQUAL WEIGHT. Answers should be justified by showing work.

Time Allowed	**Reading Time:**	*10 minutes*
	Writing Time:	*2 hours*
You are supplied with:		*No additional materials*
Calculators:		*Any basic financial calculator e.g. Texas Instruments BA 2 Plus*

Question 1: The value of a real investment is an assessment of its risks and opportunities.

a. What are the main risks that real estate investors need to take into account? [5 points]

These issues were discussed at length during Lecture 1 and then at various points during the first five weeks of the module.

b. How do these feed into your cash flow calculations and how do you assess the impact of such risks on your financial analysis? [5 points]

As above, these concepts were discussed during Lecture 1 and then at various points during the first five weeks of the module. Mostly risks affects the choice of discount rate. But "flexing" a cash flow model allows to assess the sensitivity of analysts' financial decision-making to key parameters (such as vacancies, exit yields, rental concessions, rental appreciations, etc.)

c. What are the main opportunities that arise from real estate investments? [5 points]

These issues were discussed at length during Lecture 1 and then at various points during the first five weeks of the module.

d. Rents and sale value: why are they intrinsically linked? And why should a 'smart' asset manager focus on maximising rental values? [10 points]

These issues were discussed during Lecture 1 and then at various points during the first five weeks of the module. Rents determine sale value through a cap-rate operation. Smart manager should focus on maximising rental values rather than engineering higher prices by compressing exit yields.

e. Discuss the main lessons you have learnt from working at your two case studies in terms of strategic asset management, revenue maximisation and risk mitigation. [15 points]

There is no set answer and I look forward to hearing what the main take-away lessons were from the point of view of the students when considering the Croydon and Leeds properties and different markets.

f. How do climate change considerations alter some of the perceptions about risks and opportunities in real estate investment? What mitigations/actions are 'green' real estate investors taking in relation to the down and upsides created by heightened attention to climate issues? [20 points]

This is a complex question that was discussed in the pre-recorded materials by Dr. Liao as well as the external-speaker pane discussions held in Week 9. There are no pre-set answers.

g. Real estate funds change the way in which different 'actors' (limited and general partners) are exposed to real-estate risks. Discuss some of the key features of fund structures in relation to management roles, incentives and targets, and 'waterfall' distribution of risks and revenues. [20 points]

These issues were discussed in the pre-recorded materials by Prof. Silva and then again during the external-speaker lecture of Week 7.

h. Different real-estate fund investors will have different priorities depending on their institutional features, 'culture' and country of origin. Discuss some of the key insights in these respects that were presented during the relevant external-speaker lecture? [20 points]

Damian Perry – our external speaker of Week 7 – gave a one-hour and a half lecture on these topics. Although the material was quite complex, I am expecting the students will be able to unpick the main elements and formulate their own judgment.

Question 2: You need to borrow £5,000,000 for a period of five years. You are considering loan facilities with monthly payments and compounding, and you are offered an Interest Only (IO) loan with an annual interest of 5%. There are no origination charges or discount points.

a- What is the value of the monthly payments you owe to the bank? And what is the size of the balloon payment at the end of Year 5? [5 points]

The loan interest on a monthly basis is 5%/12 and this is charged on 5,000,000. The monthly charges are therefore 20,833.33. This is over and above the reimbursement of 5,000,000 in the last 'balloon' payment.

b- You are now instead considering a constant payment mortgage (CPM). The loan also has monthly compounding and charges an annual interest of 5%. However, unlike the IO, the CPM amortizes over time and is structured to come to maturity after 10 years from origination. What is the value of the monthly instalments you need to pay to the bank? What is the outstanding balance on this loan at the end of Year 5? [10 points]

The loan stretches over 120 periods (10 years with monthly compounding) with a monthly charge of 5%/12. Students can work out the quantities using the time-value of money worksheet and setting the appropriate compounding. The amount of constant monthly payments are 53,032.7576 while the outstanding balance on this 10 year loan after 5 years (60 periods) is 2,810,243.286.

c- The bank has also informed you that both loans carry a 0.5% discount point and a 0.5% pre-payment penalty. What is the effective cost of borrowing for the IO loan if you reimburse it at the end of Year 5? And what is the effective cost of borrowing for the CPM loan if you repay at the end of Year 5? Compare your findings and discuss what you notice. [15 points]

First, the students should notice that the discount points apply to both loans but the pre-payment penalty only applies to the CPM. This last 10 years but is reimbursed in year 5. The IO loan is instead already structured over five years and does not incur a penalty.

The IO IRR or effective cost of borrowing can be easily worked out with the cash flow function of the calculator. The discount point is worth 25,000 (0.005 of 5,000,000). So vis-à-vis 5,000,000 – 25,000 (CF0), the bank is making 59 periods of 20,833.33 and one last period of 20,8333 + 5,000,000. This gives an annual IRR of 5.1135%.

The IRR of the CPM can also be calculated with the cash flow worksheet. The outflow for the bank is the same. The monthly payments are 53,032.7576 and the last payment further includes the outstanding balance at 2,810,243.286 – further increased by a pre-payment penalty at 0.005. That is: 2,824,294.502. This gives an IRR of 5.203%. Students could have similarly used the time-value of money worksheet, recognising they can work out I/Y and input FV.

d- What is the rationale for banks to impose origination charges? What do banks 'like' in a CPM loan compared to an IO loan? Why might borrowers instead prefer an IO loan? [5 points]

These issues were discussed at length during Lectures 4 and 5 of the course.

e- Consider now that you are borrowing this amount to purchase a building worth £10,000,000 that generates a monthly NOI of 62,500 throughout the five years. You also expect to sell the building at end of your holding period for £12,500,000. What is the annual IRR on this investment without leverage? [10 points]

The unleveraged IRR can be found using the cash flow worksheet with CF0=-10,000,000; then 59 periods of payments at 62,500 and one final period of 12,500,000 + 62,500. The IRR is: 11.2479%.

f- Now calculate the leveraged annual IRRs using the two loan structures discussed above (IO, and CPM) inclusive of the discount points and other charges spelled out at point c-. What do you find and what do you notice? Why do you think the loans give more/less favourable results? [25 points]

The answer to this point is just a bit more "computationally" intensive, but the students have all the required elements from point e- and the previous ones.

For the IO, the CF0 becomes 10,000,000 – 4,975,000 (5mio minus the discount points). This is an outflow of money, so it should be imputed with a negative sign. Then the 59 monthly payments will be 62,500 – 20,833.33 (the IO charges). The last payment will be 12,500,000 – 5,000,000 (sale value minus loan reimbursement) plus another 62,500 – 20,833.33. This gives an IRR of 16.3759%

Similarly, for the CPM most of the data were provided above. CF0 is once again 10,000,000 – 4,975,000. Then the 59 monthly payments will be 62,500 – 53,032.7576 (the CPM charges). The last payment will be 12,500,000 – 2,824,294.502 (sale value minus loan reimbursement inclusive of the pre-payment charge) plus another 62,500 – 53,032.7576. This gives an IRR of 14.85%.

Both loans give favourable leverage- which is expected because IRR pre-leverage (11.24%) is above the cost of borrowing. But the IO is more advantageous. This does not have much to do with the prepayment charge (for the CPM), but because of the fact that the CPM has amortization the "eats into" the value of the early cash flows. Having a lower loan to reimburse at the end of the holding period (discounted 60 periods from now) does not compensate for earlier lower amounts.

g- Now you realise your building also requires some uplifting and remodelling. This implies the following: i- during the first 6 months of your holding period, the building will be empty and so it will not generate any NOI; ii- you expect the overall costs of such remodelling to be 5% of the initial value of the building; iii- you assume these overall costs will be disbursed in equal instalments over the initial six months; and iv- during this period, you will take an additional loan facility to defray the cost of the works. This short-term (6 months) development Interest Only (IO) loan has monthly instalments and charges an annual interest of 7.5%. You agree with the bank that they will provide you with enough funds to cover the cumulative cost of such remodelling works. How much do you borrow from the bank? And what are the interest payments owed to the bank? [10 points]

These instructions imply the following. During the first six months the building will not generate 62,500. Next, the overall cost of the works is 5% of 10,000,000 or 500,000. These are to be spread across months – so it will be 500,000/6=83,333. The bank is actually providing these 500,000 which you use in order to "neutralise" the 83,333 – so no cost in practice from the works. But you are charged an annual interest at 7.5% or 0.625% per month. This applies to the 500,000 you borrow – giving a monthly cost of 3,125. At the end of the 6 months, you give the bank 500,000 back.

h- Reconsider now the overall cash-flow from this investment and focus on the main IO loan facility described above (including the additional charges at point c-). What is the annual leveraged IRR on this investment including the remodelling costs and development loan? What would instead be the IRR if you ONLY borrowed the money required to complete the remodelling works? Compare your findings and discuss what you notice. [20 points]

This follows from what was already modelled above. However, the following changes are required.

CF0 remains 10,000,000 – 4,975,000 (5mio minus the discount points). The 500,000 (the borrowing for the works) do not reduce the purchase costs but are used to get the costs of the works down to zero (and pay interests instead). Then there will be 5 periods of monthly payments with zero rent, a main loan charge of 20,833.33 and a development loan charge of 3,125. In the sixth period, the charges further include the reimbursement of the 500,000. After that, there will be 53 periods where the payments are 62,500 – 20,833.33 (the IO charges). The last payment will be 12,500,000 – 5,000,000 (sale value minus loan reimbursement) plus another 62,500 – 20,833.33. This gives an IRR of 12.5204%.

If we only take the development loan, then the figures would be as follows. CF0 10,000,000. Then there will be 5 periods of monthly payments with zero rent and a development loan charge of 3,125. In the sixth period, the charges further include the reimbursement of the 500,000. After that, there will be 53 periods where the payments are 62,500. The last payment will be 12,500,000 plus another 62,500. This gives an IRR of 9.2889%. Not only this is lower than just here above but also lower than the unleveraged finding at point e-. This is not surprising as some costs have to be disbursed in the form of interest payments in order to carry out the works.